Maximilian Rücker

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Education

Ulm University May 2023 – today
Ph.D. Mathematics
- Research topic: High-dimensional panel data models.
- Supervisor: Prof. Dr. Michael Vogt

Ulm University Mar 2021 – Mar 2023
M.Sc. Mathematics
- Thesis: "High-Dimensional Inference with the Lasso."
- Final Grade: 1.2

Ulm University Oct 2017 – Mar 2021
B.Sc. Business Mathematics
- Thesis: "Fractional Poisson Processes and Fractional Poisson Fields."
- Final Grade: 1.8

German grading scale: 1.0 is the highest grade.

Publications

Estimation and Inference in High-Dimensional Panel Data Models with Interactive Fixed Effects Nov 2025
Quantitative Economics, 2025.
Joint with Michael Vogt, Oliver Linton and Christopher Walsh.

Job Market Paper

High-Dimensional Panel Data Models with Interactive Fixed Effects: Beyond the Linear Case Aug 2026
arXiv preprint, `http://arxiv.org/abs/2608.02055`.
Joint with Michael Vogt and Oliver Linton.

Abstract. Modern economic panel data sets are often high-dimensional: they contain information on a wide variety of control variables whose number may even exceed the sample size. Nevertheless, the literature on econometric methods for high-dimensional panels is quite limited. In this paper, we study high-dimensional panel models with interactive fixed effects where the regression function has an additive structure, i.e., each covariate enters the model via an unknown nonlinear component function. We develop estimation methodology and theory in this additive framework which substantially extends previous work on the high-dimensional linear case by Rücker et al. (2025). In the theoretical part of the paper, we derive the convergence rate of our estimator for both the small-T and the large-T panel case. The theory is complemented by comprehensive Monte Carlo experiments and an empirical application.

Work in Progress

Inference in high-dimensional additive models
Joint with Michael Vogt.

Abstract. In this work, we develop a specification test for a nonlinear treatment effect in the high-dimensional additive model. We establish asymptotic validity under the null hypothesis "no treatment effect" and consistency under fixed alternatives satisfying a standard completeness condition.

Selected Talks

31st International Panel Data Conference July 2026
Speaker at the IPDC 2026 in Exeter about "High-Dimensional Panel Data Models with Interactive Fixed Effects: Beyond the Linear Case".

Cambridge Econometrics Seminar (Ph.D.) Mar 2025
Talk about "Additive High-Dimensional Panel Data Models with Interactive Fixed Effects."

29th International Panel Data Conference July 2024
Speaker at the IPDC 2024 in Orléans about "Estimation and Inference in High-Dimensional Panel Data Models with Interactive Fixed Effects".

Scientific Activities

H2O Workshop Aarhus Mar 2026
Participant and poster presentation at the H2O (Higher-Order pattern-discovery in High-dimensional data) conference in Aarhus.

Workshop organisation July 2025
Organisation of a workshop for PhD students in Mathematics at Ulm University.

University of Cambridge Mar 2025
One month research stay at the University of Cambridge supervised by Oliver Linton

Statistical Packages

R-package `hdcce` Nov 2025
Available at GitHub for the HD-CCE–estimator proposed in Rücker et al. (2025).

R-package `LassoNoiseInference` Mar 2023
Available at GitHub for the estimation of the Lasso's effective noise proposed in "Estimating the Lasso's Effective Noise" (2021) by Johannes Lederer and Michael Vogt.

Teaching Experience

Teaching assistant	Oct 2024 – present
Nonparametric statistics (Bachelor's level)	Winter term 25
Asymptotic statistics (Master's level)	Summer term 25
High-dimensional statistics (Master's level)	Winter term 24
Scientific assistant	May 2023 – Sept 2024
Student assistant	Oct 2019 – Mar 2023
Calculus II (Bachelor's level)	Winter term 22
Calculus I (Bachelor's level)	Summer term 22
Econometrics (Bachelor's level)	Winter term 21
Stochastic processes and Probability theory (Bachelor's level)	Summer term 21
Stochastic processes and Probability theory (Bachelor's level)	Summer term 20
Measure theory (Bachelor's level)	Winter term 19

Refereeing

Journal of Econometrics.

Software

Programming languages
R, Python, Java and MATLAB.
Technologies
LaTeX, GitHub, HTML and CSS.

References

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Jan Beyersmann
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